



Eventive Carpooling

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Abstract

Carpooling, also known as car-sharing, is a relatively new technology that has made travel more convenient and efficient for ordinary people. By sharing a vehicle and its associated costs, individuals can reduce their travel expenses, fuel consumption, and exhaustion. As the population grows, personal vehicles are no longer able to provide adequate transportation, leading to traffic congestion and increased air and noise pollution. Carpooling is the ideal solution to address these issues.

To facilitate carpooling, we plan to develop an Android-based application that will allow passengers to collaborate with like-minded people and plan their journeys using a user-friendly interface. Our primary goal is to offer a mobility service that enables users to plan, book, and pay for multiple transportation services through carpooling via a joint digital platform. This concept represents a shift away from personally owned modes of transportation and towards mobility provided as a service. The focus is on providing real-time solutions to travelers based on their specific travel needs. This carpooling system demonstrates how users can reach their destination by combining multiple rides.

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Introduction

This Carpooling system aims to provide one stop solution to daily commute problem without any hassle, making it more flexible and reliable while creating great connections with the help of an e-platform where commuters come together to share pleasant journey. Carpooling is a revolutionary and a unique way of travelling. We want to provide a hassle-free transit to our users along with a great experience.

TITLE : Eventive CarPooling

Illustration:



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Spine : BS(CS) | Eventive Carpooling | 2023

Blurb : Sharing Rides , Connecting Communities

Quote : Collaborate with Commuters, Cut Your Commuting Costs and Reduce Your Carbon Footprint

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Related Work

Carpoolyn [1] is Pakistan's 1st Carpooling (Ride-Sharing) App with an aim to make Traveling accessible, affordable, and safe for Everyone Across Pakistan. It connects the drivers with passengers traveling on the same routes to share the fuel cost. Have A Car? Just charge for empty seats in your car and make money to cover your commuting cost. Need A Ride? Share a ride with drivers heading towards the same destinations & reduce your traveling cost.

Ridely [2] is rideshare services in Pakistan. It's an emerging app that bring a new concept of route matching in Carpool. This carpooling app allows its users to select rides based on the route match between two cities of Pakistan.

Savaree [3] is a Pakistan-based cab service that offers affordable and safe rides. It offers a carpooling app that allows users to find rides to places they would like to go. It also matches passengers with drivers going along specific routes or to specific events. Savaree was founded in 2014 and is based in Lahore, Pakistan. It also provides inter-city services.

BlaBlaCar [4] is the world's largest long-distance ridesharing community. Conceived in December 2003 by Frédéric Mazzella, and founded in 2006, BlaBlaCar connects drivers and passengers willing to travel together between cities and share the cost of the journey. BlaBlaCar has more than 20 million members across 19 countries. Members must register and create a personal online profile, which includes ratings and reviews by other members, social members show how much experience they have of the service, meaning those with more-known as "ambassadors" - attract more ride shares.

FolksVagn [5] offers a community-based system that helps people share rides with others. While the passengers get rides at costs much cheaper than a taxi



service, the car owner gets a share of the fare. It is open only to corporate clients as it requires a corporate email for registration and has a prepaid account or online wallet system to pay for the ride.

Requirements

Hardware Interfaces

The application will be hosted. API will be used for location/gps functionality. The mobile application for User and Client can be installed on any mobile device which runs the latest Android versions or latest IOS version.

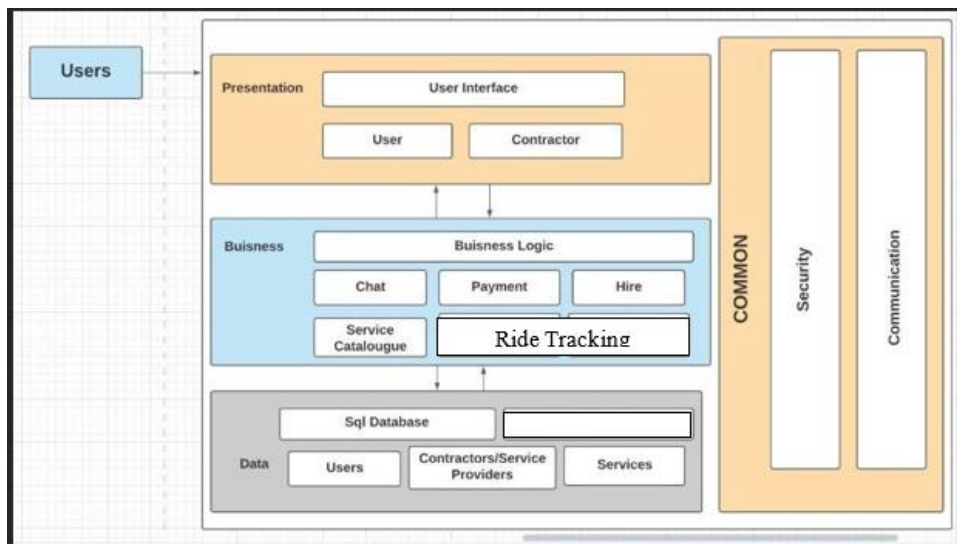
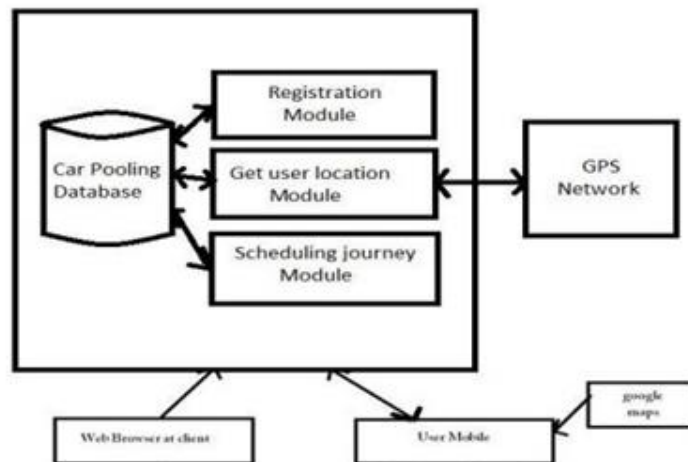
Software Interfaces

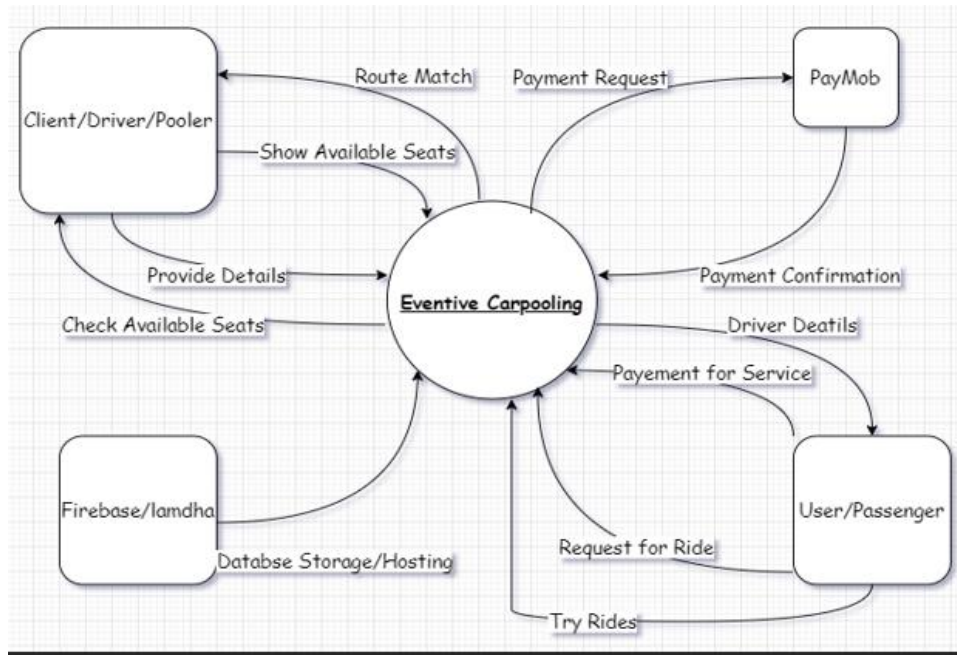
Application is developed using React Native. Firebase is used for database management. Through online payment feature is provided. Web application of Admin is not an operating system restricted as any system with browsers such as Chrome, Edge or Firefox can run the application. The mobile application of User and Client can be installed on the latest versions of Android or IOS.

Communications Interfaces

App will communicate with the user by showing possible rides and request rides. Payment details will be encrypted and secured. Password will be encrypted with Firebase Authentication which uses an internally modified version of script to hash account passwords. Even when an account is uploaded with a password using a different algorithm.

Designs





Tables

	Chat	Intra-City Sharing	Inter-City Sharing	Gender Specific	Realtime Connectivity	Road Statistics based fare
Savaree	✓	✓	✗	✗	✗	✗
Ridely	✓	✗	✓	✗	✗	✗
Carpoolyn	✓	✗	✗	✗	✗	✗
Careem	✓	✗	✗	✗	✗	✗
Uber	✓	✗	✗	✗	✗	✗
InDrive	✓	✗	✗	✗	✗	✗
Eventive	✓	✓	✓	✓	✓	✓

Table 1 : Comparative Analysis Table

Implementation

Carpooling system provides the benefits of sharing rides with different peoples. By surveying different papers, it is found that it uses heuristic searching techniques for searching the cars. By using Euclidean distance techniques, it finds the nearest car that has to be assigned to the passenger. It usually uses Global Positioning System (GPS) for finding the locations of cars. In study it is found that, Route matching

techniques are used in order to match drivers and passenger's locations by using Google maps API, and Then based on their route, the driver is allocated to the passenger. In most carpooling systems, it is observed that the Dijkstra algorithm is used for detecting the shortest route of destination for a passenger. The Dijkstra algorithm is usually used for detecting the optimal minimum path. It is found that it checks for maximum possibilities and then chooses the minimum route for the journey. Similarly, Dijkstra algorithm is also used to connect to the nearest vehicle.

Testing and Evaluation

Testing and evaluation are critical steps in ensuring the functionality, usability, and user experience of a carpooling app. Here are some steps for testing and evaluation:

1. **Functionality testing:** This testing ensures that the app is functioning correctly and without errors. Testers should try different features and scenarios to verify that they work as intended. For example, testers can check if the app can match drivers and passengers, book rides, and provide directions.
2. **Usability testing:** This testing focuses on the ease of use and user experience of the app. Testers should evaluate the app's user interface, navigation, and how easy it is for users to accomplish their goals. For example, testers can evaluate the clarity of the app's language, the placement of buttons, and how easy it is to create and join rides.
3. **Compatibility testing:** This testing ensures that the app works seamlessly across different devices and platforms. Testers should try the app on different devices, including smartphones, tablets, and desktops, with different operating systems and browsers. For example, testers can check if the app is compatible with iOS and Android devices and works on different screen sizes.
4. **Performance testing:** This testing evaluates the app's performance under different conditions. Testers should check how quickly the app loads and responds, how well it performs under different network conditions, and how well it handles large amounts of data. For example, testers can evaluate the app's response time, its ability to handle multiple users simultaneously, and its resilience under network interruptions.
5. **Security testing:** This testing ensures that the app is secure and protects user data. Testers should verify that the app uses secure protocols for data transfer and storage, and that it has appropriate authentication and access controls. For example, testers can evaluate if the app encrypts user data, if it has strong password policies, and if it protects against SQL injection attacks.
6. **User feedback and evaluation:** This evaluation involves getting feedback from real users of the app. Testers should conduct surveys or user interviews to gather feedback on the app's functionality, usability, and user experience. This feedback can then be used to identify areas for improvement and make changes to the app.

In summary, testing and evaluation are critical steps in ensuring the

functionality, usability, and user experience of a carpooling app. By conducting thorough testing and evaluation, developers can identify issues and improve the app's performance, security, and user experience.

Conclusion

In conclusion, a carpooling app can provide numerous benefits, including reducing traffic congestion, lowering transportation costs, and reducing carbon emissions. By leveraging technology, carpooling apps can connect drivers and passengers, making it easier to find rides and reduce the number of single-occupancy vehicles on the road. Additionally, carpooling apps can create a sense of community and help users build social connections.

To ensure the success of a carpooling app, it is essential to focus on user experience, functionality, and security. Thorough testing and evaluation can help identify areas for improvement and ensure that the app meets the needs of its users.

Overall, carpooling apps have the potential to create a more sustainable, cost-effective, and enjoyable transportation experience for commuters. As more people become aware of the benefits of carpooling, we can expect to see an increase in the adoption of carpooling apps and a reduction in traffic congestion, air pollution, and transportation costs.

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